

ARC Figures

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```
library(tidyverse)
library(dplyr)
library(readxl)
library(ggplot2)
library(ggtext)
library(stringr)
```

```
frequencies_renamed <- read_excel("Frequencies - PI and Cxn to CPS.xlsx") |>
  mutate(denominator = recode_values(denominator,
                                     "imprisoned people" ~ "Imprisoned People",
                                     "incarcerated parents" ~ "Incarcerated Parents",
                                     "foster care children" ~ "Foster Care Children",
                                     "CPS-involved children" ~ "CPS Involved-Children"
                                     ))
```

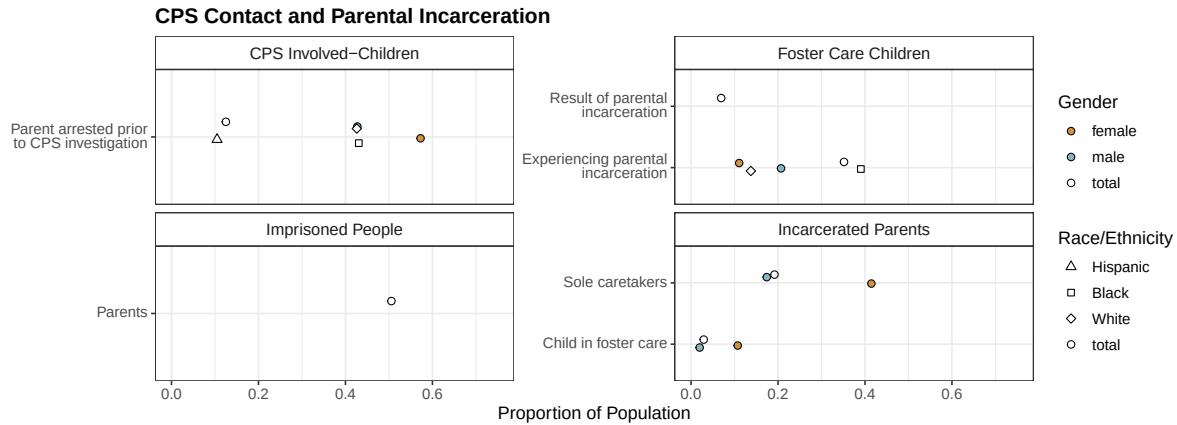
```
frequencies_pivoted <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, White, Black, Hispanic),
               names_to = "demographic",
               values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("White", "Black", "Hispanic") ~ demographic,
    TRUE ~ "total"
  ),
  gender = case_when(demographic %in% c("male", "female") ~ demographic,
                     TRUE ~ "total"),
  cps_cj = if_else(denominator %in% c("Incarcerated Parents",
                                     "Imprisoned People"),
                  "Criminal Justice Contact",
                  "Child Protective Services Contact"),
  race = fct_relevel(race, "Hispanic", "Black", "White", "total")) |>
```

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select(-demographic)

frequencies_pivoted |>
  ggplot(aes(x = proportion, y = clean_name,
            fill = gender, shape = race)) +
  geom_point(size = 2,
            position = position_jitter(height = .15),
            color = "Black",
            show.legend = TRUE) +
  scale_y_discrete(labels = c("Parent arrested prior to CPS investigation" =
                             "Parent arrested prior<br>to CPS investigation",
                             "Result of parental incarceration" =
                             "Result of parental<br>incarceration",
                             "Experiencing parental incarceration" =
                             "Experiencing parental<br>incarceration")) +
  facet_wrap(~denominator, scales = "free_y") +
  scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
                              "total" = "white")) +
  scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,
                              "White" = 23, "total" = 21)) +
  theme_bw() +
  theme(strip.background = element_rect(fill = NA),
        axis.title.y = element_blank(),
        strip.text.x = element_text(
          face = "plain",
          size = 11,
          margin = margin(t = 6, b = 6)
        ),
        plot.title = element_text(face = "bold"),
        text = element_text(size = 12),
        axis.text.y = element_markdown(),
        plot.caption.position = "plot",
        plot.caption = element_markdown(hjust = 0)) +
  guides(fill = guide_legend(
    override.aes = list(shape = 21, color = "Black"))) +
  xlim(0, 0.75) +
  labs(title = "CPS Contact and Parental Incarceration",
       x = "Proportion of Population",
       y = "Condition",
       shape = "Race/Ethnicity",
       fill = "Gender")

```



```
frequencies_mutated <- frequencies_renamed |>
  pivot_longer(cols = c(total, male, female, White, Black, Hispanic),
               names_to = "demographic",
               values_to = "proportion") |>
  filter(!is.na(proportion)) |>
  mutate(race = case_when(
    demographic %in% c("White", "Black", "Hispanic") ~ demographic,
    TRUE ~ "total"),
    gender = case_when(demographic %in% c("male", "female") ~ demographic,
                      TRUE ~ "total"),
    alt_denominator = case_when(
      str_detect(clean_name, "parental incarceration") ~ "Foster care and CPS-involved children",
      str_detect(clean_name, "Parental incarceration") ~ "Foster care and CPS-involved children",
      str_detect(clean_name, "Parent arrested") ~ "Foster care and CPS-involved children",
      TRUE ~ "Incarcerated people"),
    race = fct_relevel(race, "Hispanic", "Black", "White", "total")
  ) |>
  select(-demographic)

frequencies_mutated |>
  ggplot(aes(x = proportion, y = clean_name,
             fill = gender, shape = race)) +
  geom_point(size = 2,
             position = position_jitter(height = .15),
             color = "Black",
             show.legend = TRUE) +
  scale_y_discrete(labels = c("Parent arrested prior to CPS investigation" =
                             "Parent arrested prior<br>to CPS investigation",
                             "Result of parental incarceration" =
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      "Result of parental<br>incarceration",
      "Experiencing parental incarceration" =
      "Experiencing parental<br>incarceration")) +
facet_wrap(~alt_denominator, scales = "free_y") +
scale_fill_manual(values = c("male" = "#86B6C3", "female" = "#D29141",
      "total" = "White")) +
scale_shape_manual(values = c("Black" = 22, "Hispanic" = 24,
      "White" = 23, "total" = 21)) +
theme_bw() +
theme(strip.background = element_rect(fill = NA),
      axis.title.y = element_blank(),
      strip.text.x = element_text(
        face = "plain",
        size = 11,
        margin = margin(t = 6, b = 6)
      ),
      plot.title = element_text(face = "bold"),
      text = element_text(size = 14),
      axis.text.y = element_markdown(),
      plot.caption.position = "plot",
      plot.caption = element_markdown(hjust = 0)) +
guides(fill = guide_legend(
      override.aes = list(shape = 21, color = "Black"))) +
xlim(0, 0.75) +
labs(title = "CPS Contact and Parental Incarceration",
      x = "Proportion of Population",
      y = "Condition",
      shape = "Race/Ethnicity",
      fill = "Gender")

```

